

Linear Algebra and Calculus Lecture 6

Eric Sandin Vidal

Fall 2025

Vectors

A **column vector** (or simply **vector**) is a matrix with only one column. For example

$$\begin{bmatrix} 5 \\ 6 \end{bmatrix}, \begin{bmatrix} \sqrt{2} \\ \frac{1}{3} \end{bmatrix}, \begin{bmatrix} \pi \\ 1 \end{bmatrix}.$$

We will write $\mathbb{R}^2$ for the set of 2×1 matrices or column vectors with two (real number) entries. Elements will typically be written as $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$. Note that the addition of vectors is properly defined:

$$v + w = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \end{bmatrix},$$

which can be represented by the parallelogram rule for addition. It is also defined the multiplication by a scalar c :

$$c \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} c \cdot v_1 \\ c \cdot v_2 \end{bmatrix}.$$

More generally for an integer $n \geq 1$ we can consider the set $\mathbb{R}^n$ of $n \times 1$ matrices with real numbers as entries. As before there is an addition on $\mathbb{R}^n$ and a scalar multiplication $c \cdot v$ for $v \in \mathbb{R}^n$ and c a scalar. These satisfy the following algebraic properties

- | | |
|--------------------------------|-------------------------|
| 1. $u + v = v + u$ | 5. $c(u + v) = cu + cv$ |
| 2. $u + (v + w) = (u + v) + w$ | 6. $(c + d)u = cu + du$ |
| 3. $v + 0 = 0 + v = v$ | 7. $c(du) = (cd)u$ |
| 4. $u + (-u) = -u + u = 0$ | 8. $1u = u$ |

Linear combinations

Let $v_1, \dots, v_p$ be elements of $\mathbb{R}^n$. Then we say an element $y \in \mathbb{R}^n$ is a **linear combination** of $v_1, \dots, v_p$ if there are scalars called **weights** $c_1, \dots, c_p \in \mathbb{R}$ such that $y = c_1 v_1 + \dots + c_p v_p$. For example,

$$\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Question 1. Let $a_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$, $a_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$ and $b = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$. Can b be generated (or written) as a linear combination of a_1 and a_2 ?

This amounts to finding the weights (if they exist) of such linear combination. That is, we want to find real numbers x_1, x_2 such that

$$x_1 a_1 + x_2 a_2 = b.$$

If we write the values of the vectors we see

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

which is the same as

$$\begin{bmatrix} x_1 \\ -2x_1 \\ -5x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 5x_2 \\ 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

and

$$\begin{bmatrix} x_1 + 2x_2 \\ -2x_1 + 5x_2 \\ -5x_1 + 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}.$$

Two vectors are equal if their entries are equal, so this holds true if

$$\begin{aligned} x_1 + 2x_2 &= 7 \\ -2x_1 + 5x_2 &= 4 \\ -5x_1 + 6x_2 &= -3 \end{aligned}$$

Which is a linear system of equations! We can write the augmented matrix and row reduce it:

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 9 & 18 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

The solution is $x_1 = 3$ and $x_2 = 2$. Hence b is a linear combination of a_1 and a_2 .

Exercise. Plug the solution of x_1 and x_2 into the linear combination equation and check that the equality holds.

Proposition. A vector equation

$$x_1 a_1 + x_2 a_2 + \cdots + x_n a_n = b$$

has the same solution set as the linear system whose augmented matrix is

$$\begin{bmatrix} a_1 & a_2 & \cdots & a_n & b \end{bmatrix}.$$

In particular, b can be generated by a linear combination of $a_1, \dots, a_n$ if and only if there exists a solution to the linear system corresponding to the previous matrix.

Span

Definition. The **span**, $\text{Span}\{v_1, \dots, v_p\}$, of a set of vectors $v_1, \dots, v_p \in \mathbb{R}^n$ is the set of vectors $w \in \mathbb{R}^n$ that are a linear combination of $v_1, \dots, v_p$, that is, that can be written in the form

$$c_1v_1 + c_2v_2 + \dots + c_pv_p$$

with $c_1, \dots, c_p$ scalars.

Asking whether a vector b is in $\text{Span}\{v_1, \dots, v_p\}$ amounts to asking whether the vector equation

$$x_1v_1 + \dots + x_pv_p = b$$

has a solution, or, equivalently, asking whether the linear system with augmented matrix $\begin{bmatrix} v_1 & \dots & v_p & b \end{bmatrix}$ has a solution.

The Matrix Equation $Ax = b$

Recalling the previous lecture, it is possible to linearly combine the columns of a matrix. Let us formalize that.

Definition. If A is an $m \times n$ matrix, with columns $a_1, \dots, a_n$, and if $x \in \mathbb{R}^n$, then the product of A and x , denoted by Ax , is the linear combination of the columns of A using the corresponding entries in x as weights; that is

$$Ax = \begin{bmatrix} a_1 & a_2 & \dots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1a_1 + x_2a_2 + \dots + x_na_n.$$

Example. For $v_1, v_2, v_3 \in \mathbb{R}^n$, we can write the linear combination $3v_1 - 5v_2 + 7v_3$ as a matrix times a vector:

$$3v_1 - 5v_2 + 7v_3 = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix} = Ax.$$

The equation $Ax = b$ is called the **matrix equation**.

Theorem. If A is an $m \times n$ matrix, with columns $a_1, \dots, a_n$ and if $b \in \mathbb{R}^m$, the matrix equation

$$Ax = b$$

has the same solution set as the vector equation

$$x_1a_1 + \dots + x_na_n = b$$

which in turn, has the same solution set as the system of linear equations whose augmented matrix is

$$\begin{bmatrix} a_1 & a_2 & \dots & a_n & b \end{bmatrix}.$$

It follows from this theorem, that the equation $Ax = b$ has a solution if and only if b is a linear combination of the columns of A .

Example. Let $A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Is the equation $Ax = b$ consistent for all possible b_1, b_2, b_3 ?

We can row reduce the augmented matrix for $Ax = b$:

$$\begin{bmatrix} 1 & 3 & 4 & b_1 \\ -4 & 2 & -6 & b_2 \\ -3 & -2 & -7 & b_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 7 & 5 & b_3 + 3b_1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 0 & 0 & b_3 + 3b_1 - \frac{1}{2}(b_2 + 4b_1) \end{bmatrix}.$$

The third entry in column 4 equals $b_1 - \frac{1}{2}b_2 + b_3$. The equation $Ax = b$ is not consistent for every b because some choices of b can make $b_1 - \frac{1}{2}b_2 + b_3$ non-zero. For it to be consistent it is required that

$$b_1 - \frac{1}{2}b_2 + b_3 = 0.$$

Theorem. Let A be an $m \times n$ matrix. Then the following statements are logically equivalent. That is, for a particular A , either they are all true statements or they are all false.

1. For each $b \in \mathbb{R}^m$, the equation $Ax = b$ has a solution.
2. Each $b \in \mathbb{R}^m$ is a linear combination of the columns of A .
3. The columns of A span $\mathbb{R}^m$.
4. A has a pivot position in every row.

Exercise. Compute the matrix-vector product using the row-vector rule for the following matrix and vector:

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix}.$$

Theorem. If A is an $m \times n$ matrix, u and v are vectors in $\mathbb{R}^n$, and c is a scalar, then:

1. $A(u + v) = Au + Av$;
2. $A(cu) = c(Au)$.